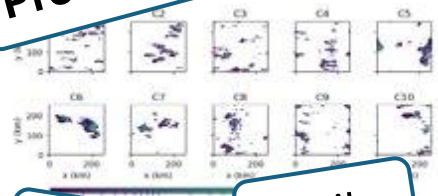
How should we parametrize MCSs?

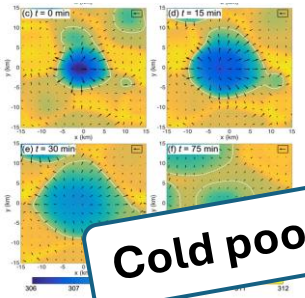
Mark Muetzelfeldt, University of Reading

Process models



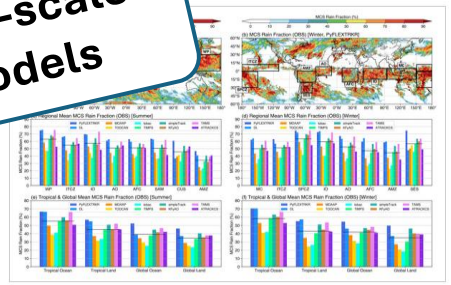
convection

anvils

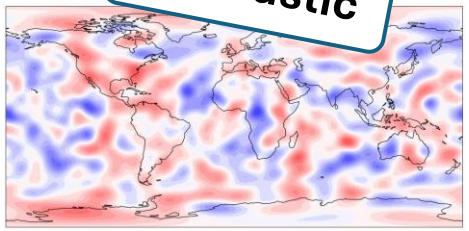


Cold pools

km-scale models



stochastic



Existing schemes

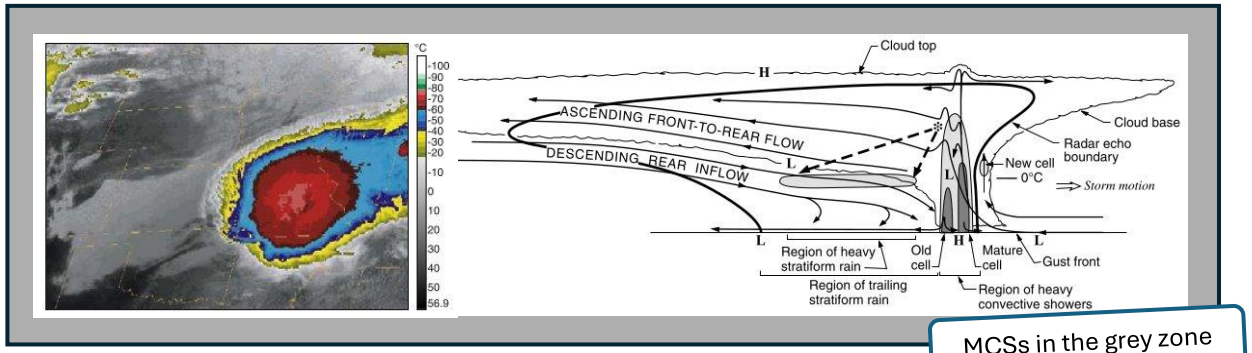
- Prognostic entrainment
- Multiscale Coherent Structure Parametrization (MCSP)
- Stochastic MultiCloud Model (SMCM)
- Cold pool/org
- Cellular automata
- Superparametrization
- …

Evidence base

mesoscale circulation

Scale break

Length scales

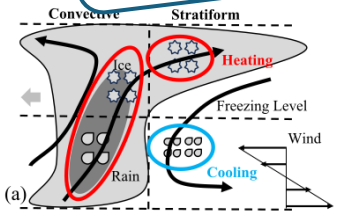


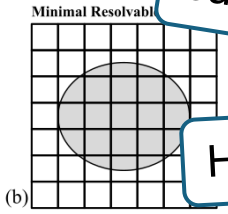
MCSs in the grey zone

What is the grey zone for MCSs?

resolution

subgrid/surgrid





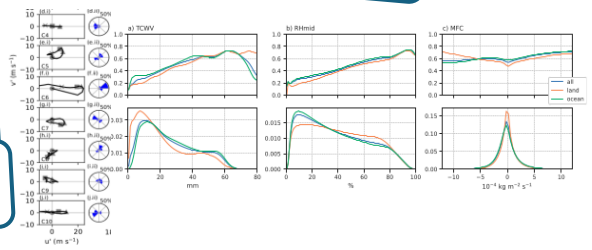
Hard/soft GZ

Environmental controls

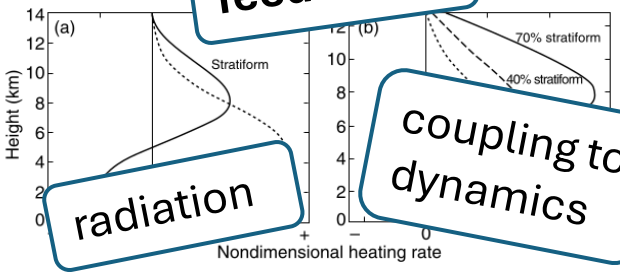
shear

TCWV

MFC



feedbacks



radiation

coupling to dynamics